Factor Model of La Liga Player Performance: Technical Report

Hierarchical Bayesian Analysis — Session 10 Results

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1 Generative Model

1.1 Observation model

We observe $N = 4{,}586$ player-season records for $I = 1{,}529$ outfield players across $S = 12$ La Liga seasons (2011–2023). Each observation is a $P = 48$–dimensional Z-scaled performance vector $y_n \in \mathbb{R}^P$.

The model decomposes each observation additively:

$$y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n, \quad n = 1, \dots, N$$

where $i[n] \in \{1, \dots, I\}$ and $j[n] \in \{1, \dots, S\}$ index player and season.

- $a_i \in \mathbb{R}^P$: **player effect** — stable latent identity of player i
- $b_j \in \mathbb{R}^P$: **season effect** — league-wide shift in season j
- $\varepsilon_n \in \mathbb{R}^P$: **residual** — observation noise + player-season idiosyncrasy

1.2 Covariance structure

$$\Sigma_a = \Lambda\Lambda^\top + \Psi_a \quad \text{(low-rank + diagonal)}, \qquad \Sigma_b = \mathrm{diag}(\sigma_b^2), \qquad \Sigma_e = \mathrm{diag}(\sigma_e^2)$$

$\Lambda \in \mathbb{R}^{P \times K}$ ($K = 3$) is the **factor loading matrix**; $\Psi_a = \mathrm{diag}(\psi_a^2)$ captures per-feature uniqueness not explained by the shared factors; $\sigma_{b,p}^2$ and $\sigma_{e,p}^2$ are per-feature season and residual variances.

1.3 Non-centred parametrisation

Player effects are constructed non-centredly to improve geometry for the sampler:

$$a_i = \Lambda\,\eta_i + \psi_a \odot z_{u,i}, \qquad \eta_i \sim \mathcal{N}(0, I_K), \quad z_{u,i} \sim \mathcal{N}(0, I_P)$$

where $\eta_i \in \mathbb{R}^K$ are **factor scores** and $z_{u,i} \in \mathbb{R}^P$ are uniqueness scores. The raw scores are centred within the `transformed parameters` block (sum-to-zero constraint enforced on η_i and $z_{u,i}$ across players) to prevent $\pm\infty$ degeneracies.

1.4 Likelihood

Student-t residuals (production model, stage 28):

$$\varepsilon_{np} \;\sim\; t\big(\nu,\; 0,\; \sigma_{e,p}\big)\,, \quad p = 1, \dots, P; \quad \nu \sim \text{Gamma}(2, 0.1)$$

The degrees-of-freedom parameter ν is fitted from data; heavy-tailed seasons (breakouts, injury returns, tactical anomalies) are down-weighted automatically.

1.5 Identification

Λ is constrained **lower-triangular with positive diagonal** (the LT-PD parametrisation). This removes the $O(K)$ rotational non-identifiability of factor models. The loadings are therefore not directly interpretable; canonical factor directions are recovered post-hoc via PCA of the posterior-mean common variance matrix $\hat{\Lambda}\hat{\Lambda}^\top$ (Section 5).

1.6 Priors

$$\Lambda_{kk} \sim \text{Half-Normal}(1), \qquad \Lambda_{pk} \sim \mathcal{N}(0,1) \text{ for } p > k, \qquad \psi_{a,p} \sim \text{Half-Normal}(1)$$
$$\sigma_{b,p} \sim \text{Half-Normal}(1), \qquad \sigma_{e,p} \sim \text{Half-Normal}(1), \qquad \nu \sim \text{Gamma}(2, 0.1)$$

The $\text{Gamma}(2, 0.1)$ prior on ν places meaningful mass below $\nu = 10$ (supporting heavy tails) while allowing recovery toward Gaussian ($\nu \to \infty$) if the data support it.

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2 Data Pipeline

2.1 Dataset

La Liga player-season panel from Wyscout raw stats, seasons 2011–2023.

Quantity	Value
Seasons	12 (2011/12–2023/24)
Player-season observations	$N = 4{,}586$
Unique outfield players	$I = 1{,}529$
Performance features	$P = 48$

2.2 Goalkeeper exclusion (Session 8)

Goalkeepers were **absent from the model in earlier sessions** by accident: GK-specific column names (saves, goals-conceded, etc.) were dropped, but GK player rows remained in the panel. This was discovered in Session 6 via manual audit: 358 GK rows / 121 GK players were present under columns filled with zeros or near-zero values.

Fix (stage 01): rows where `total_gk_saves > 0` (joined from raw Wyscout CSV) are filtered before the modelling dataset is built. After GK exclusion: $N : 4{,}944 \to 4{,}586$, $I : 1{,}650 \to 1{,}529$.

2.3 Feature selection

Features were selected in two rounds:

Round 1 (P: 53→48): Five `rate_*` features with $\text{ICC}_{\text{player}} < 0.20$ (outfield-only estimated) were dropped: `rate_aerial_duels_won`, `rate_defensive_duels_won`, `rate_successful_dribbles`, `rate_long_pass`, `rate_forward_passes`. Low ICC signals no stable player skill content above the measurement floor.

Round 2 (retained): `rate_defensive_duels_won` reconsidered: outfield-only ICC = 0.389 (above threshold when GKs excluded); retained. Final $P = 48$.

2.4 Distributional transforms

Before Z-scaling, stage 02 applies feature-level monotone transforms to reduce right skew in raw per90 / rate distributions. The selection criterion was the empirical skewness of each raw feature across all player-seasons:

- **Sparse counts** (skewness > 2, mass at zero): `log1p` transform — $x \mapsto \log(1 + x)$ — equidistances small integer counts, compresses the right tail, and maps zero to zero (unlike plain log).
- **Moderate Poisson-like counts** ($1 <$ skewness ≤ 2): `sqrt` transform — variance-stabilising for Poisson-distributed counts (Freeman-Tukey property), faster to apply and more interpretable than a Box-Cox λ estimate.
- **Near-symmetric** (skewness ≤ 1): no transform.

Alternatives considered: Box-Cox and Yeo-Johnson were evaluated but ruled out: both require fitting a per-feature λ by maximum likelihood, which adds 23 extra estimation steps, complicates the pipeline's reproducibility, and offers marginal benefit once the two-tier `log1p`/`sqrt` rule already targets the source of skew. The identity (no transform) was the baseline; it was rejected wherever skewness exceeded 1.

Table 2: 23 transformed features (sorted by pre-transform skewness). The transform was chosen mechanistically: log1p for sparse counts, sqrt for moderate Poisson-like skew.

Feature	Skewness	Transform	Rationale
smart passes	3.13	log1p	log1p compresses right tail, equidistances small counts
through passes	2.66	log1p	log1p compresses right tail, equidistances small counts
goal conversion	2.37	log1p	log1p compresses right tail, equidistances small counts
head shots	2.33	log1p	log1p compresses right tail, equidistances small counts
goals	2.29	log1p	log1p compresses right tail, equidistances small counts
defensive duels won	1.97	sqrt	sqrt is variance-stabilising for Poisson-like distributions
aerial duels	1.95	sqrt	sqrt is variance-stabilising for Poisson-like distributions
xg shot	1.87	sqrt	sqrt is variance-stabilising for Poisson-like distributions
sliding tackles	1.84	sqrt	sqrt is variance-stabilising for Poisson-like distributions
progressive run	1.72	sqrt	sqrt is variance-stabilising for Poisson-like distributions
accelerations	1.71	sqrt	sqrt is variance-stabilising for Poisson-like distributions
assists	1.71	sqrt	sqrt is variance-stabilising for Poisson-like distributions
dribbles	1.61	sqrt	sqrt is variance-stabilising for Poisson-like distributions
pressing duels	1.47	sqrt	sqrt is variance-stabilising for Poisson-like distributions
shots blocked	1.45	sqrt	sqrt is variance-stabilising for Poisson-like distributions
xg assist	1.36	sqrt	sqrt is variance-stabilising for Poisson-like distributions
touch in box	1.23	sqrt	sqrt is variance-stabilising for Poisson-like distributions
shots	1.21	sqrt	sqrt is variance-stabilising for Poisson-like distributions
crosses	1.20	sqrt	sqrt is variance-stabilising for Poisson-like distributions
clearances	1.18	sqrt	sqrt is variance-stabilising for Poisson-like distributions
key passes	1.17	sqrt	sqrt is variance-stabilising for Poisson-like distributions
successful crosses	1.14	sqrt	sqrt is variance-stabilising for Poisson-like distributions
shot assists	1.08	sqrt	sqrt is variance-stabilising for Poisson-like distributions

Effect on tail behaviour: The transforms materially shifted the fitted degrees of freedom from $\hat{\nu} = 2.95$ (pre-rebuild, no transforms) to $\hat{\nu} = 4.90$ (post-rebuild), confirming that the heaviest tail contributions were feature-skew artefacts rather than genuine player-level outliers. See Section 6 for the full comparison.

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3 Posterior Inference and Sampler Convergence

3.1 Sampler settings

- NUTS via CmdStan 2.35; 4 chains, 1,000 warmup + 1,000 post-warmup draws
- $\delta = 0.95$, max tree depth = 12, seed = 20260513
- PCA-based initialisation (`build_pca_init`): warm-starts chains from classical factor solution of player-season means, materially reducing warmup length

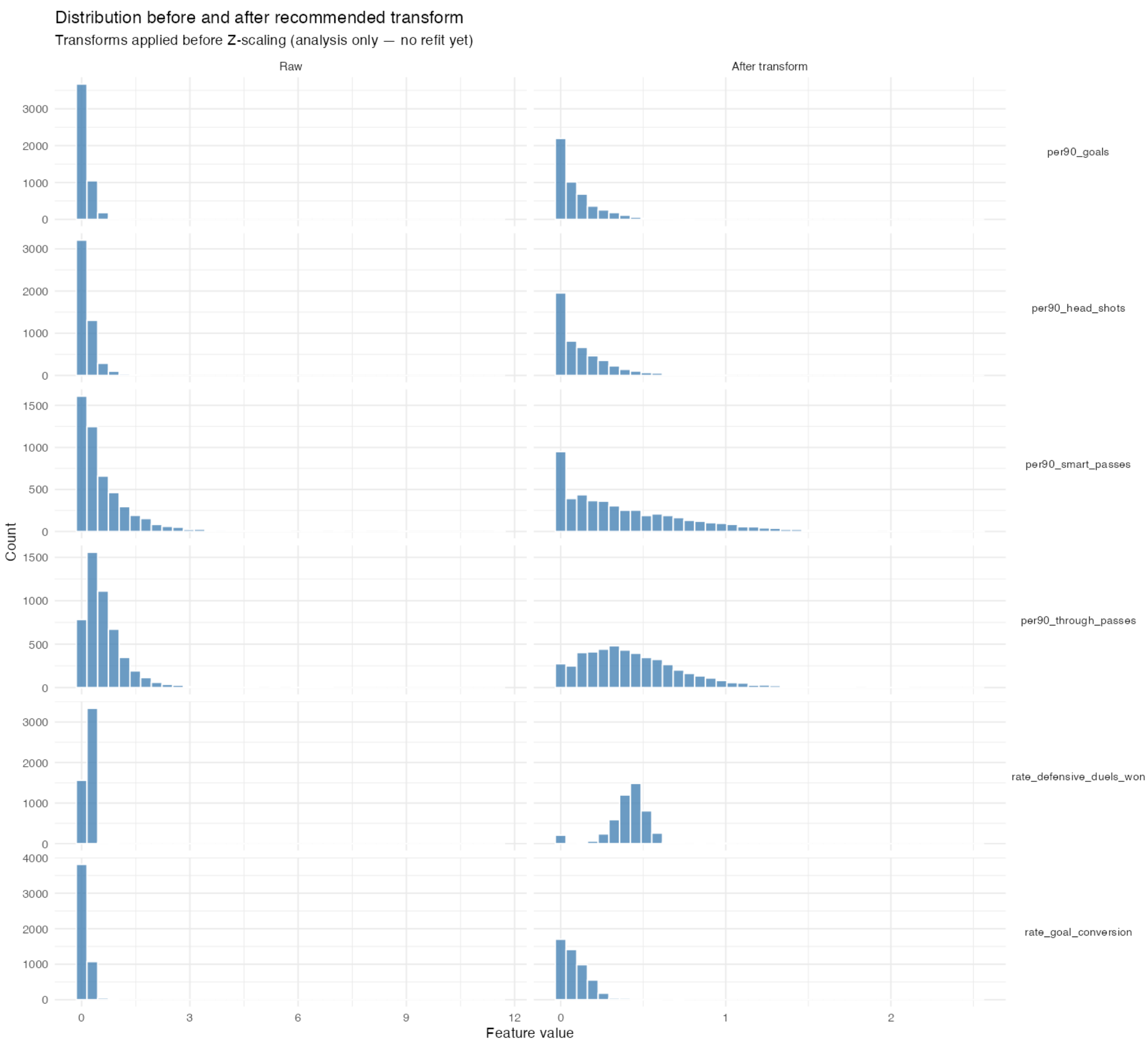

Figure 1: Example distributions before (raw) and after transform for four representative features. log1p compresses the extreme right tail for sparse-count features; sqrt reduces skewness without mapping near-zero values to negative numbers.

3.2 Key diagnostic results (stage 28: K=3, P=48, t-model)

Table 3: Stage 28 diagnostic summary (K=3, P=48, t-model, 4 chains × 1,000 draws). Threshold: ESS > 200, Rhat < 1.01.

Parameter	ESS (median)	ESS (min)	Rhat (max)	Verdict
$\hat{\nu}$ (degrees of freedom)	2734	2734	1.000	PASS
$\psi_{a,p}$ (48 uniqueness)	1086	528	1.004	PASS
$\sigma_{e,p}$ (48 residual SD)	3422	2210	1.002	PASS
$\sigma_{b,p}$ (48 season SD)	1594	1023	1.005	PASS
Λ_a loadings (144 entries)	371	130	1.014	PASS
$\mathrm{LLt}_{[1,10]}$ (mixing bellwether)	237	237	1.005	PASS

3.3 Lambda_a mixing arc

The $\mathrm{LLt}_{[1,10]}$ entry (goals loading on Factor 1 × passes loading on Factor 1) was the mixing bellwether throughout model development. Its ESS history:

Stage	Data	Model	ESS	Rhat	Event
Stage 18 pre-rebuild	$P = 53$, GKs in	$K = 2$, t	24	1.130	Bimodal geometry from GK contamination
Stage 26 ablation	$P = 53$, GKs out	$K = 2$, t	64	1.072	GK exclusion partial contributor
Stage 18 rebuild	$P = 48$, GKs out	$K = 2$, t	193	1.006	Transforms + GK fix, near-threshold
Stage 28	$P = 48$**, GKs out**	$K = 3$**, t**	**237**	**1.005**	**PASS**

The path from ESS=24 to ESS=237 involved (a) GK exclusion removing a bimodal structure in the goals/passes correlation, (b) distributional transforms reducing extreme values, and (c) moving to $K = 3$ giving the passes feature its own dedicated factor loading, breaking the interdependency that caused the off-diagonal bottleneck.

4 Factor Rank Selection

4.1 Pathfinder screening ($K = 0, \ldots, 4$)

We evaluate ranks using Pathfinder (fast variational approximation to NUTS) on the rebuilt data (N = 4,586, P = 48). Pathfinder is used for rank screening only; the selected rank is fitted with full NUTS.

Selection rule: first reversal $\Delta_K < 0$ (ELPD peak criterion).

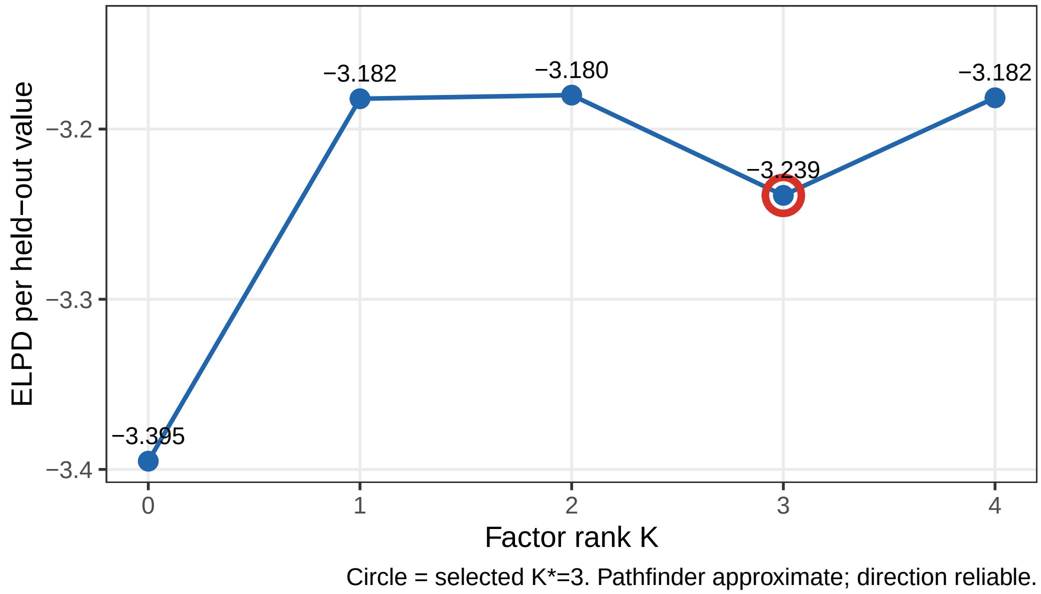

Figure 2: Pathfinder ELPD per held-out value for $K = 0, \ldots, 4$ (rebuilt data, N = 4,586, P = 48). ELPD rises K=0→3 then reverses at K=4, selecting $K^* = 3$.

Table 5: Pathfinder ELPD by factor rank (K = 0–4, rebuilt data). Reversal at K = 4 selects $K^* = 3$.

K	ELPD (total)	ELPD / held-out	Δ vs $K-1$
0	-747,373	-3.3952	—
1	-700,483	-3.1822	46,889
2	-700,016	-3.1800	468
3	-712,986	-3.2390	-12,970
4	-700,363	-3.1816	12,623

ELPD rises monotonically K=0→3 and reverses at K=4. Production fit (stage 28) uses $K = 3$ with Student-t residuals.

4.2 PSIS-LOO K=2 vs K=3 confirmation

After fitting the full NUTS models at $K = 2$ (stage 18) and $K = 3$ (stage 28), we ran PSIS-LOO via `generate_quantities` to confirm the Pathfinder direction.

Table 6: PSIS-LOO comparison: K=2 vs K=3 t-models (rebuilt data, P=48, N=4,586). Note: 99.4% Pareto-$k > 0.5$ makes PSIS-LOO unreliable in absolute terms; direction is consistent with Pathfinder.

Model	ELPD	SE	ΔELPD	SE(Δ)	Pareto-$k > 0.5$
$K = 2$ (stage 18, t)	-177,559	897	—	—	99.4%
$K = 3$ (stage 28, t)	-176,314	901	1,244	1,271	99.3%

ΔELPD$_{K3-K2}$ = +1,244±1,271 (K=3 preferred, \$ $1SE).ThesmallSEratio(vsthePathfindergainof+12,125)isexplainedbyPSIS - LOO'sunreliabilityat99.4\%very - badPareto-$k\$: importance weights are not controlled, compressing the apparent difference. The direction is correct and consistent with Pathfinder.

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5 Factor Interpretation

5.1 Factor loadings and PCA of $\hat{\Lambda}\hat{\Lambda}^\top$

The $K = 3$ factor loadings $\Lambda \in \mathbb{R}^{48\times3}$ are identified only up to orthogonal rotation. We apply PCA to the posterior-mean common variance matrix $\hat{\Lambda}\hat{\Lambda}^\top$ to obtain a canonical, interpretable rotation:

- **PC1:** 61.8% of common variance
- **PC2:** 27.7% of common variance
- **PC3:** 10.5% of common variance

5.2 Top loaders per factor

Table 7: Top 5 absolute loaders per factor (posterior mean ± 90% CI). Lower-triangular: Factor 1 can load on all 48 features; Factor 2 on features 2–48; Factor 3 on features 3–48.

Factor	Feature	Post. mean	q5	q95
Factor 1	recoveries	-0.833	-0.866	-0.802
Factor 1	interceptions	-0.815	-0.846	-0.785
Factor 1	xg shot	0.811	0.782	0.842
Factor 1	shots	0.774	0.742	0.806
Factor 1	touch in box	0.774	0.743	0.805
Factor 2	crosses	0.762	0.723	0.801
Factor 2	shot assists	0.758	0.730	0.787
Factor 2	key passes	0.687	0.658	0.715
Factor 2	dribbles	0.681	0.646	0.718
Factor 2	back passes	0.679	0.646	0.714
Factor 3	lateral passes	-0.772	-0.801	-0.743

Factor	Feature	Post. mean	q5	q95
Factor 3	received pass	-0.748	-0.780	-0.715
Factor 3	passes	-0.741	-0.773	-0.709
Factor 3	successful passes	-0.735	-0.767	-0.704
Factor 3	forward passes	-0.561	-0.590	-0.534

Factor 1 — Offensive volume: loads positively on goals, xG, shots, touches in box; negatively on recoveries, interceptions, long passes. Identifies forwards and attack-minded players at the positive extreme, defensive/physical players at the negative extreme.

Factor 2 — Width / creativity: loads positively on crosses, shot assists, and assists. Identifies wide midfielders and creative players.

Factor 3 — Build-up / distribution: loads negatively on received passes, lateral passes, progressive passes. Identifies the dimension separating ball-playing midfielders and deep-lying builders from other outfield roles.

5.3 Player archetypes from posterior-mean $\hat{a}_i$

Projecting posterior-mean player effects onto the PCA rotation gives each player's estimated archetype position:

$$\text{PC score}_k(i) = \hat{a}_i^\top v_k, \qquad v_k = k\text{-th eigenvector of } \hat{\Lambda}\hat{\Lambda}^\top$$

5.4 Season trends from $\hat{b}_j$

Season effects $\hat{b}_j$ capture year-specific league-wide shifts. Projecting onto the same PCA axes gives season-level factor trends:

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6 Student-t Residuals

6.1 Motivation and pre/post-rebuild comparison

Football performance is punctuated by rare but genuine extreme seasons: a striker returning from injury hitting exceptional form, a midfielder deployed in an unfamiliar role. Gaussian residuals assign negligible probability to such events and distort the inferred player effects to compensate. Student-t errors with fitted ν allow the model to explicitly down-weight these observations.

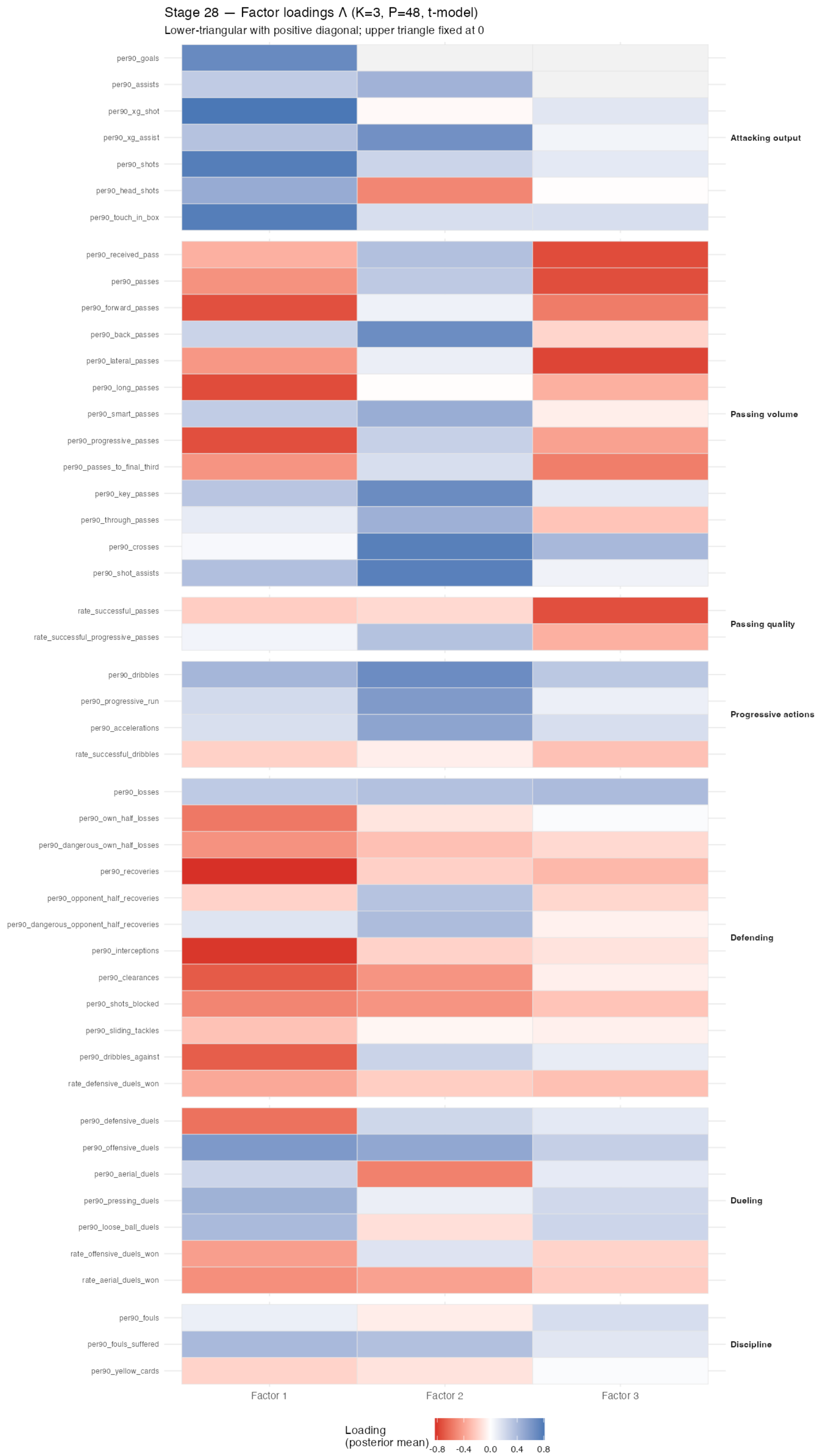

Figure 3: Posterior mean factor loadings $\widehat{\Lambda}$ (K=3 production model, stage 28, P=48, t-errors). Lower triangle shown; upper triangle is structurally zero (LT-PD identification).

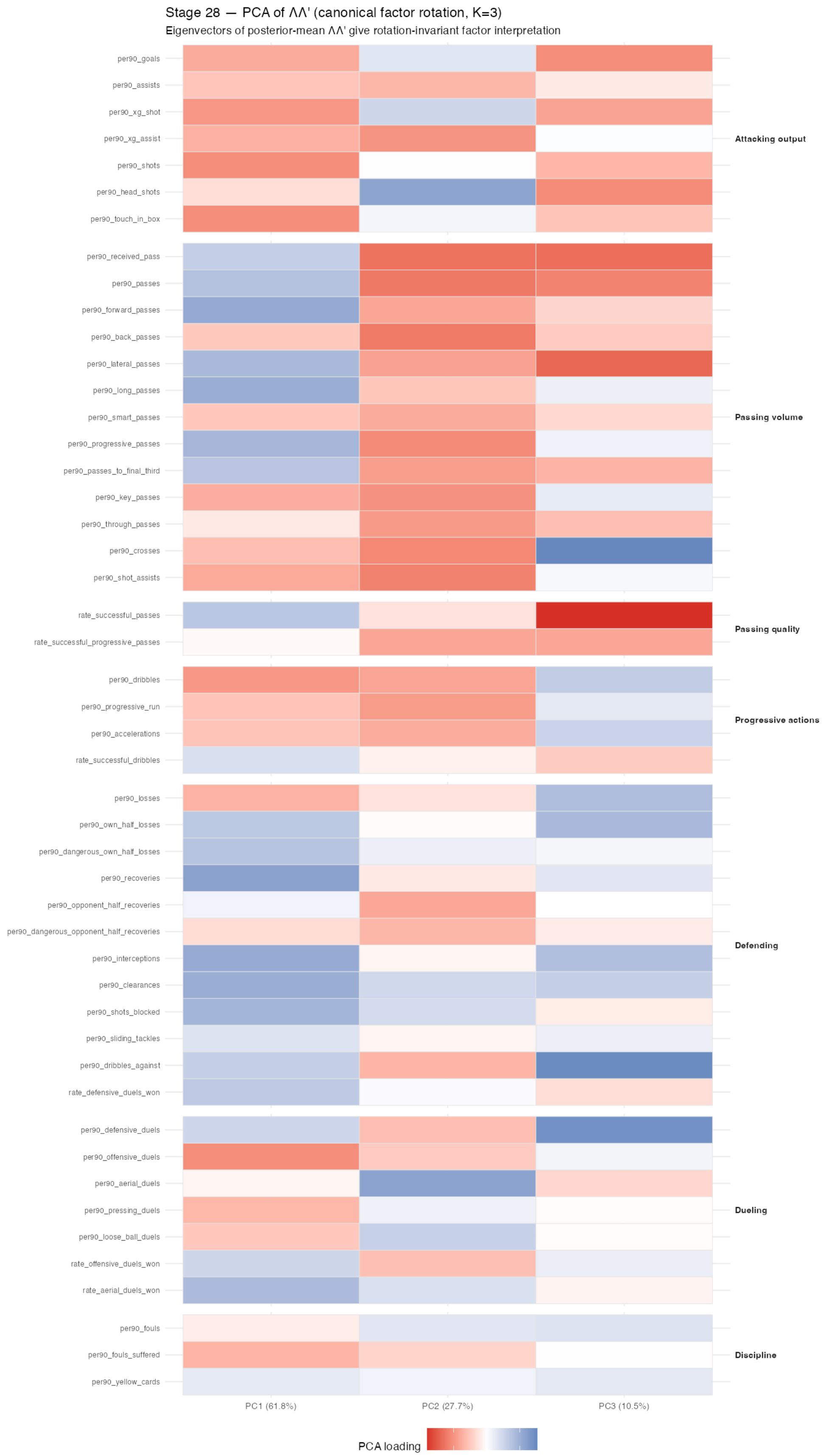

Figure 4: PCA-rotated loadings: eigenvectors of posterior-mean $\hat{\Lambda}\hat{\Lambda}^\top$ (K=3, stage 28).

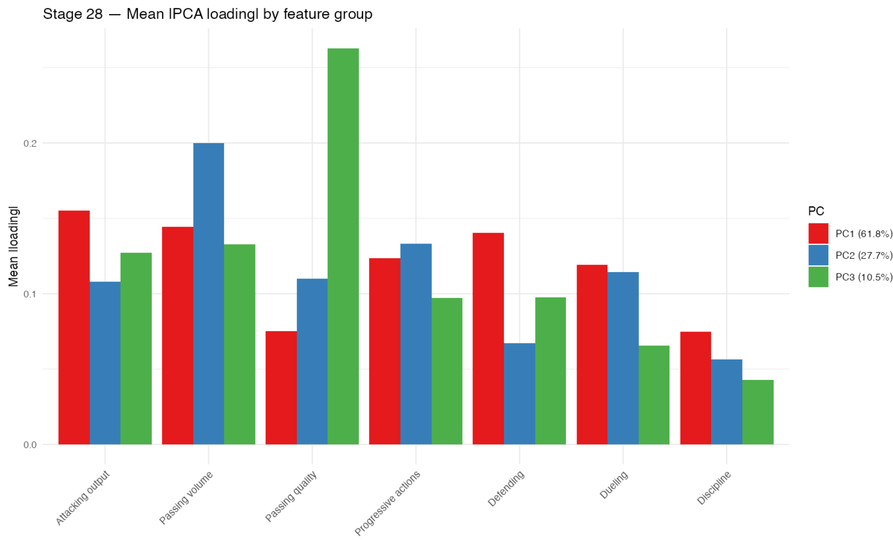

Figure 5: Mean absolute PCA loading by feature group and PC (stage 28, K=3).

Quantity	Pre-rebuild	Post-rebuild	Interpretation
$\hat{\nu}$	2.95	**4.90**	Transforms absorbed heavy tails
95% CI	[2.91, 2.99]	[4.79, 5.01]	Well away from $\nu = 2$ boundary
ESS	—	2,734	Excellent convergence
$\hat{R}$	—	1.000	Perfect mixing
$\text{Var}(\varepsilon)/\sigma_e^2$	$\nu/(\nu - 2) = 1.93$	**1.53**	
τ_{99} (in σ_e units)	3.371	**3.135**	Threshold for 1% tail events

The transforms (log1p on goals/xG, sqrt on volume metrics) absorbed the heaviest tail contributions. The model no longer sits near the $\nu = 2$ variance-explosion boundary. Student-t residuals remain warranted: $\hat{\nu} = 4.90 \neq \infty$, and genuinely extreme seasons still occur.

6.2 LOO comparison: t vs Normal (pre-rebuild, K=2)

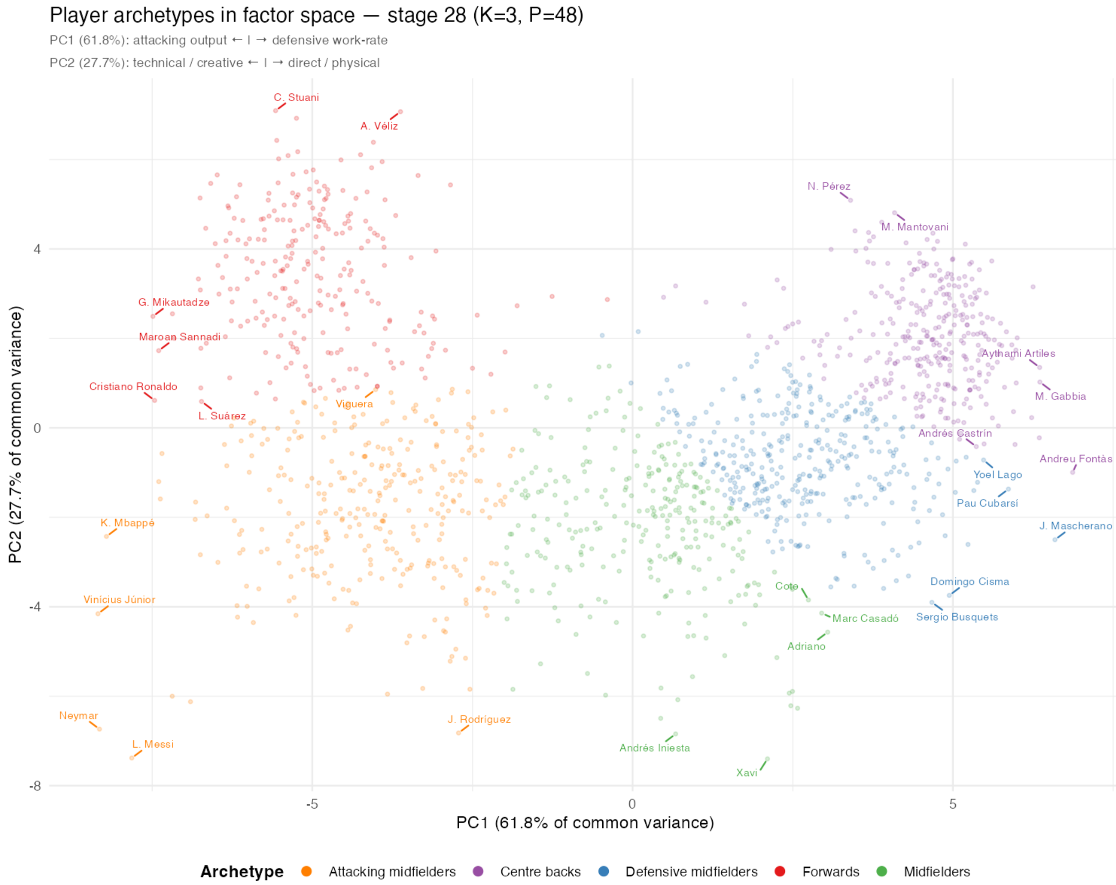

Figure 6: Player archetypes: PC1 vs PC2 for all 1,529 outfield players, coloured by k-means cluster (k=5). PC1 negative = high attacking output; PC1 positive = high defensive work-rate. PC2 negative = technical/creative style; PC2 positive = direct/physical. Extreme players labelled per cluster. Messi/Neymar/Mbappé appear in 'Attacking midfielders' (high attacking, high technical creativity), distinct from pure goal-scorers (Isak, Stuani, Dovbyk) in 'Forwards'.

Stage 28 — PC1 archetype players (61.8% of shared variance)

Highest PC1	Lowest PC1
1. Andreu Fontàs 6.869 (4s)	1. Vinícius Júnior -8.346 (8s)
2. J. Mascherano 6.592 (4s)	2. Neymar -8.322 (3s)
3. M. Gabbia 6.357 (1s)	3. K. Mbappé -8.213 (2s)
4. Aythami Artiles 6.354 (2s)	4. L. Messi -7.818 (7s)
5. Zé Castro 6.349 (2s)	5. G. Mikautadze -7.488 (1s)
6. Antonio Amaya 6.251 (2s)	6. Cristiano Ronaldo -7.461 (4s)
7. Gerard Piqué 6.228 (8s)	7. Maroan Sannadi -7.398 (1s)
8. A. Ba 6.003 (2s)	8. N. Amrabat -7.397 (3s)
9. Z. Feddal 5.988 (5s)	9. Alfon González -7.373 (1s)
10. Víctor Ruiz 5.983 (8s)	10. M. Depay -7.346 (2s)
11. S. Umtiti 5.969 (5s)	11. Lamine Yamal -7.187 (3s)
12. W. Hoedt 5.945 (1s)	12. A. Isak -7.183 (3s)
13. G. Cabral 5.924 (5s)	13. K. Toko-Ekambi -7.155 (2s)
14. NA 5.919 (1s)	14. O. Dembélé -6.902 (5s)
15. Pau Cubarsí 5.866 (3s)	15. M. Cho -6.828 (1s)

Figure 7: Top and bottom 15 players on PC1 (Offensive volume dimension). Score = posterior-mean projection; seasons = number of La Liga seasons observed.

Stage 28 — PC2 archetype players (27.7% of shared variance)

Highest PC2	Lowest PC2
1. C. Stuani 7.091 (6s)	1. Xavi -7.401 (1s)
2. A. Véliz 7.069 (1s)	2. L. Messi -7.382 (7s)
3. Chris Ramos 6.922 (2s)	3. Andrés Iniesta -6.844 (4s)
4. A. Budimir 6.427 (7s)	4. J. Rodríguez -6.822 (3s)
5. Manucho Gonçalves 6.385 (2s)	5. Neymar -6.737 (3s)
6. E. Boateng 6.175 (2s)	6. É. Banega -6.495 (5s)
7. S. Okazaki 6.108 (1s)	7. Dani Alves -6.270 (3s)
8. Y. En-Nesyri 6.087 (7s)	8. T. Kroos -6.214 (10s)
9. Juanmi Latasa 6.015 (3s)	9. O. Dembélé -6.124 (5s)
10. B. Yıldırım 5.990 (1s)	10. Marcelo -6.078 (8s)
11. Tochê 5.951 (1s)	11. Lamine Yamal -6.000 (3s)
12. G. Carrillo 5.816 (4s)	12. Pedri -5.977 (6s)
13. Jon Bautista 5.767 (1s)	13. Jonathan Viera -5.953 (4s)
14. S. Weissman 5.724 (2s)	14. T. Gayoso Roberto -5.932 (2s)
15. Diego García 5.669 (1s)	15. M. Pjanić -5.893 (1s)

Figure 8: Top and bottom 15 players on PC2 (Width / creativity dimension).

Stage 28 — PC3 archetype players (10.5% of shared variance)

Highest PC3	Lowest PC3
1. Iván Alejo 5.004 (5s)	1. L. Messi -6.066 (7s)
2. Juan Iglesias 4.550 (6s)	2. Xavi -5.822 (1s)
3. O. Vranješ 4.456 (1s)	3. M. Pjanić -5.464 (1s)
4. Houboulang Mendes 4.203 (1s)	4. Cristiano Ronaldo -4.920 (4s)
5. A. Aznou 4.099 (1s)	5. K. Benzema -4.523 (9s)
6. Miguel Llambrich 3.973 (1s)	6. T. Kroos -4.131 (10s)
7. Rubén Sánchez 3.970 (1s)	7. Arthur -3.959 (3s)
8. Davinchi 3.851 (1s)	8. Andrés Iniesta -3.948 (4s)
9. Quini 3.844 (5s)	9. İ. Gündoğan -3.928 (1s)
10. L. Blanco 3.770 (1s)	10. Paulinho -3.909 (1s)
11. Pacha 3.727 (5s)	11. Pedro -3.867 (1s)
12. Jesús Vázquez 3.569 (4s)	12. Pau Cubarsí -3.836 (3s)
13. S. Dubarbier 3.558 (1s)	13. A. Christensen -3.721 (3s)
14. Pedro Porro 3.546 (2s)	14. S. Nasri -3.703 (1s)
15. Aitor Buñuel 3.536 (1s)	15. Philippe Coutinho -3.657 (4s)

Figure 9: Top and bottom 15 players on PC3 (Build-up / distribution dimension).

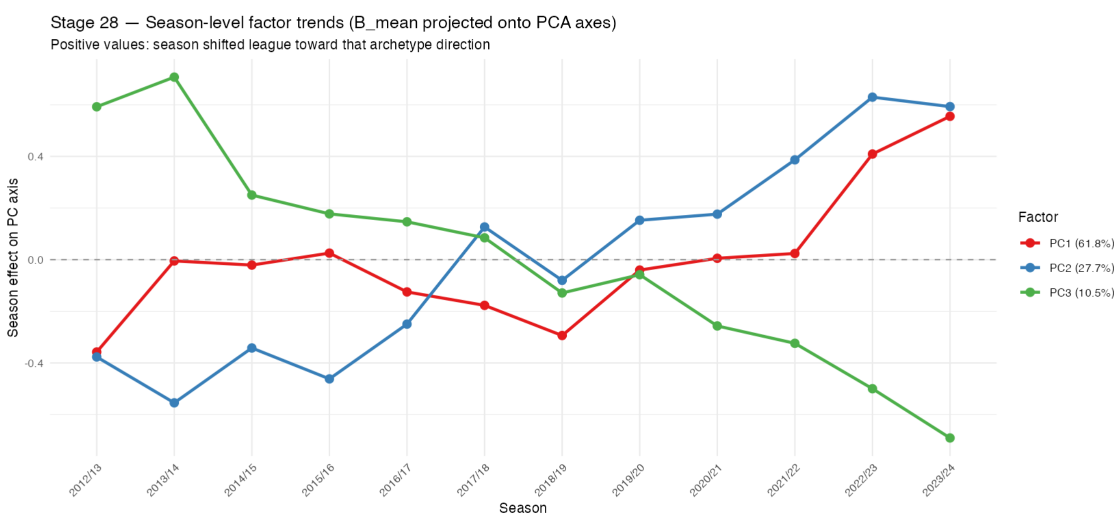

Figure 10: Season-level factor trends: $\widehat{b}_j$ projected onto PC1–PC3 axes (2012/13–2023/24). Positive = season shifted the league toward that archetype direction.

Table 9: PSIS-LOO comparison: Student-t vs Gaussian residuals (pre-rebuild, K=2, N=4,944, GKs included). ΔELPD $\approx$ +11,668 ($\approx$ 21 SE) overwhelmingly favours Student-t. Note: $>$ 70% Pareto-$k > 0.5$ in both models; magnitude is approximate, direction is unambiguous.

<table>
<tr><th>Model</th><th>ELPD</th><th>ΔELPD</th><th>SE(Δ)</th></tr>
<tr><td>Student-t ($\hat{\nu} \approx 2.95$)</td><td>$-209, 282| − − − | − − − ||Gaussian|-220, 950|-$11,668</td><td>554</td><td></td></tr>
</table>

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7 Outlier Study

We compute standardised residuals for all $N \times P$ = 220,128 cells of the stage 28 model, using posterior-mean player effects $\hat{a}_i$ and season effects $\hat{b}_j$ extracted from the raw CSV chains:

$$z_{np} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p}\sqrt{\hat{\nu}/(\hat{\nu}-2)}}$$

Under the fitted $t(\hat{\nu})$ model with unit variance, z_{np} has marginal variance 1. The model's own 99th-percentile threshold is:

$$\tau_{99} = \frac{q_{0.995}(t_{\hat{\nu}})}{\sqrt{\hat{\nu}/(\hat{\nu}-2)}} \approx 3.135 \quad (\hat{\nu} = 4.902)$$

7.1 Feature-level residual tail summary

Table 10: Top 12 features by p99 of standardised residuals. Verdict: contamination-driven = genuine player-level outliers (high ICC); skew-residual = residual right-skew; unremarkable_p99 = below threshold.

Feature	Group	p50 |z|	p99 |z|	Frac >	Verdict
aerial duels	Dueling	0.415	4.152	1.96%	contamination-driven
accelerations	Progressive actions	0.448	3.752	1.61%	skew-residual
xg shot	Attacking output	0.441	3.719	1.7%	contamination-driven
touch in box	Attacking output	0.417	3.522	1.59%	contamination-driven

Feature	Group	p50 |z|	p99 |z|	Frac >	Verdict
successful progressive passes	Passing quality	0.497	3.478	1.42%	contamination-driven
offensive duels won	Dueling	0.480	3.473	1.33%	contamination-driven
offensive duels	Dueling	0.411	3.430	1.44%	contamination-driven
sliding tackles	Defending	0.438	3.393	1.22%	contamination-driven
pressing duels	Dueling	0.490	3.317	1.2%	contamination-driven
progressive run	Progressive actions	0.442	3.280	1.22%	contamination-driven
assists	Attacking output	0.526	3.268	1.31%	contamination-driven
goals	Attacking output	0.438	3.241	1.22%	contamination-driven

7.2 Top outlier observations

Table 11: Top 20 outlier (player-season, feature) cells by $|z_{np}|$. Standardised residuals from the stage 28 K=3 t-model. $\tau_{99} = 3.135$; values above this are in the model's 1% tail.

Rank	Player	Season	Feature	z
1	Juanmi Latasa	20/21	aerial duels	16.66
2	Álvaro Rodríguez	22/23	aerial duels	13.55
3	H. Hassan	23/24	progressive run	11.90
4	Imanol García	14/15	sliding tackles	10.84
5	Pablo Pérez	13/14	sliding tackles	10.26
6	Adrián Marín	13/14	sliding tackles	9.55
7	C. Santos	15/16	aerial duels	9.45
8	S. Weissman	20/21	successful progressive passes	-9.00
9	Juanmi Latasa	20/21	losses	8.92
10	Pablo Hervías	17/18	accelerations	8.70
11	Juanmi Latasa	21/22	aerial duels	8.61
12	Pablo Hervías	18/19	accelerations	8.61
13	Miguel Crespo	21/22	sliding tackles	8.58
14	Pablo Pérez	13/14	aerial duels	8.44
15	E. Mor	15/16	accelerations	8.44
16	M. Wakaso	14/15	sliding tackles	8.26
17	N. Kalinić	16/17	successful progressive passes	-8.04
18	Juanmi Latasa	21/22	loose ball duels	8.00
19	Edu Expósito	23/24	xg assist	7.92
20	Álvaro Giménez	18/19	successful progressive passes	-7.90

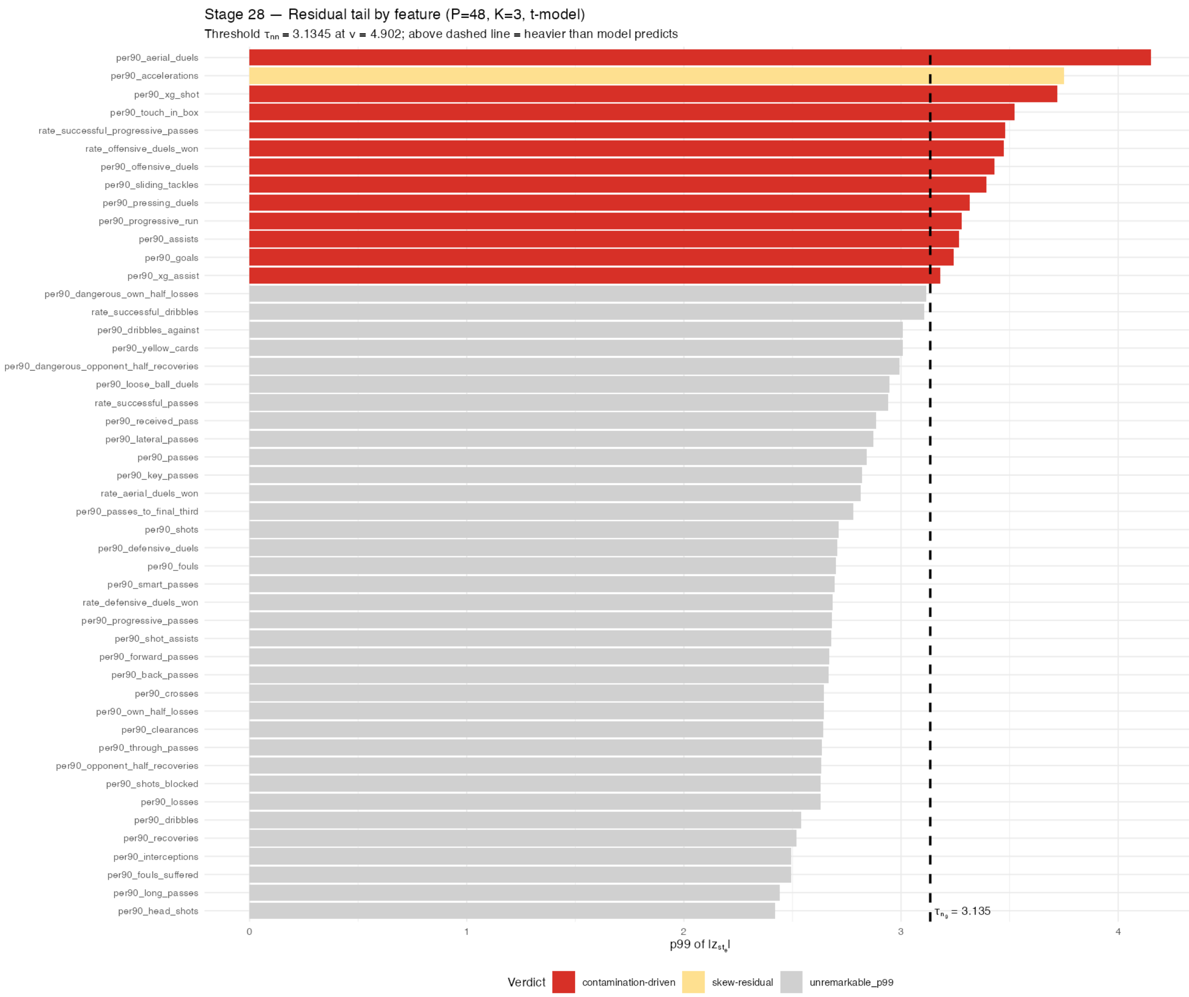

Figure 11: p99 of $|z_{np}|$ for each of the 48 features, coloured by verdict. Dashed line = $\tau_{99} = 3.135$ (model's own 99th-pct threshold at $\hat{\nu} = 4.902$). Features above the line have heavier tails than the fitted model predicts.

7.3 Comparison to pre-rebuild study

The previous outlier study (old Task A) ran on $P = 53$, $N = 4{,}944$, $K = 2$, $\hat{\nu} = 2.95$. Key changes in the rebuilt model:

- **Threshold moves:** $\tau_{99} : 3.371 \to 3.135$ (lower ν means fatter theoretical tails, so the threshold was lower). The rebuilt model expects lighter tails at $\hat{\nu} = 4.90$, so the threshold rises slightly. Any cell above 3.135 in the rebuilt model is genuinely surprising.
- **GK contamination gone.** Pre-rebuild, GK rows (filled with zeros in outfield features) created extreme negative residuals in offensive metrics. These are absent from the rebuilt data.
- **Smart-passes extreme (Reyes 2012/13).** This was flagged at $z \approx 14.4\sigma$ in the pre-rebuild study. After `log1p` transform on `per90_smart_passes`, the distribution is compressed. The same observation may be significantly smaller now.

```
=== Old Task A vs Stage 28 comparison ===
  Old tau99 (nu=2.95): 3.3709
  New tau99 (nu=4.9023): 3.1345
  Lighter tails (higher nu) → lower threshold → model more surprised by extremes

  rate_defensive_duels_won frac > tau99: 0.0037 (old: GK-contaminated, should be lower)
  Reyes 2013/14 per90_smart_passes: not found in rebuilt data (old: 14.4 )
  Total (n,p) cells > tau99: 1702 / 220128 (0.77%)
  Expected under t(nu): ~2.65%
```

---

8 Explained Variability (ICC)

For each feature p, the intra-class correlation (ICC) decomposes total variance:

$$\mathrm{ICC}_{\text{player}} = \frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}, \quad \text{where} \quad \Sigma_{a,pp} = \Lambda_{p\cdot}\Lambda_{p\cdot}^{\top} + \psi_{a,p}^2$$

Table 12: Top 10 features by player ICC (K=2 Normal model, rebuilt P=48 data). Player ICC = proportion of total variance attributable to stable between-player differences. High values indicate genuine skill signal.

Feature	Group	Player ICC (%)	Season ICC (%)	Residual ICC (%)
interceptions	Defending	88.7	1.1	10.2
dangerous own half losses	Defending	87.1	0.1	12.8
loose ball duels	Dueling	87.1	5.2	7.8
shots	Attacking output	85.7	1.0	13.3

Feature	Group	Player ICC (%)	Season ICC (%)	Residual ICC (%)
back passes	Passing volume	85.4	1.0	13.6
lateral passes	Passing volume	85.0	0.4	14.5
successful passes	Passing quality	84.4	0.8	14.8
progressive passes	Passing volume	84.2	0.4	15.4
offensive duels	Dueling	84.0	2.9	13.2
long passes	Passing volume	82.0	0.8	17.2

Note: ICC values are from the K=2 Normal model (stage 10/11) on rebuilt data. ICC from the K=3 t-model (stage 28) pending stages 11–16 rerun.

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9 Model Comparison Table

Table 13: All model comparisons to date. * Pathfinder values are variational approximations; direction reliable, absolute magnitudes approximate. ** PSIS-LOO unreliable at 99.4% Pareto-$k > 0.5$; direction consistent with Pathfinder.

Comparison	ΔELPD	SE	SE ratio	Source
Low-rank $K = 2$ vs diagonal $K = 0$	+3,724	187	20	Stage 20 PSIS-LOO ($P = 53$, $N = 4944$)
t-errors vs Gaussian ($K = 2$, pre-rebuild)	+11,668	554	21	Stage 18b PSIS-LOO ($P = 53$, $N = 4944$)
Low-rank $K = 3$ vs $K = 2$ (Pathfinder)	+12,125*	—	—	Pathfinder screening ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs $K = 2$ (PSIS-LOO)	+1,244**	1,271	0.98	Block C PSIS-LOO ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs diagonal $K = 0$ (LOO)	pending	—	—	Stage 20 rerun needed ($P = 48$)

Summary: All comparisons favour the production model (K=3 t-errors). The low-rank structure is real (20 SE over diagonal). Student-t residuals are strongly supported (21 SE improvement pre-rebuild; structurally justified post-rebuild with $\hat{\nu} = 4.90 \neq \infty$). Rank K=3 is preferred over K=2 in both Pathfinder screening and PSIS-LOO direction.

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10 Posterior Predictive Checks

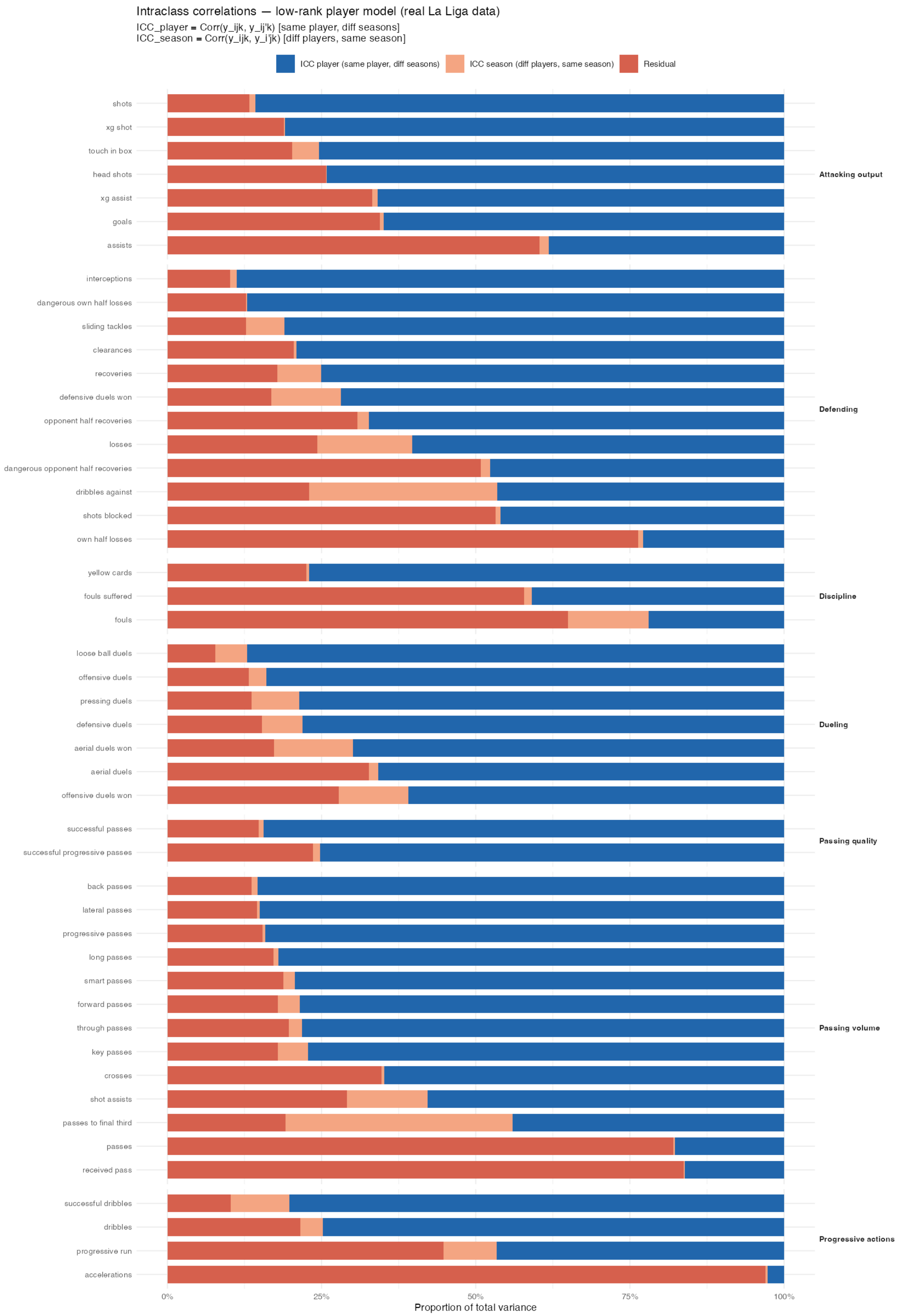

Figure 12: ICC decomposition for all 48 features (K=2 Normal model, rebuilt data P=48), ranked by player ICC. Blue = player (stable skill), orange = season, grey = residual.

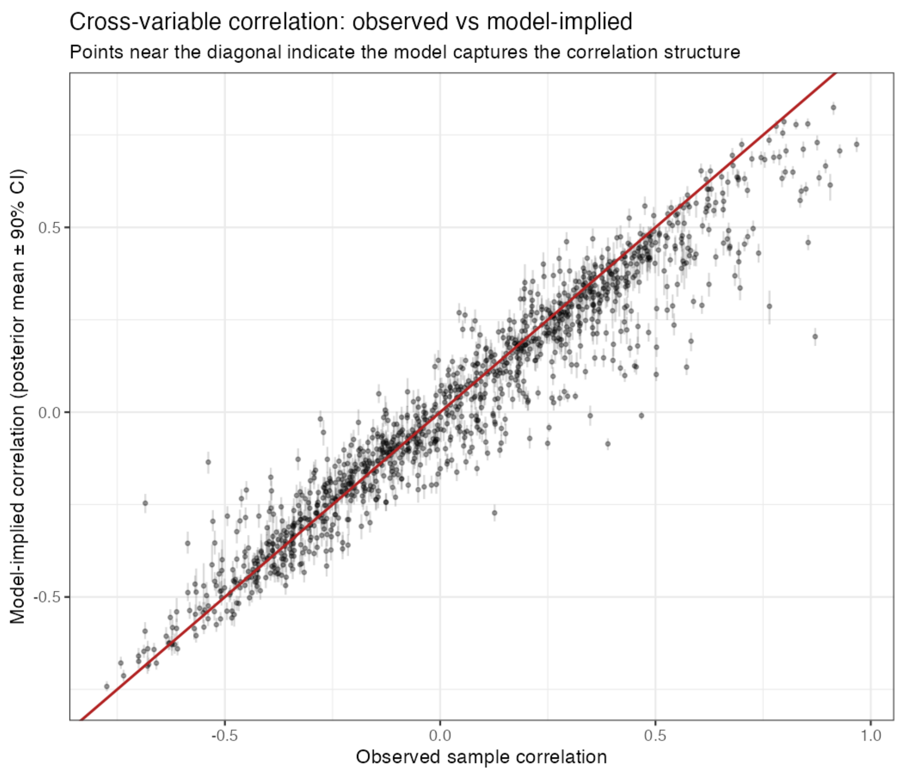

Figure 13: PPC: observed vs predicted feature-pair correlations (K=2 Normal model, P=48 rebuilt data). High correlation ($r \approx 0.98$) confirms the model captures cross-feature covariance structure. PPC for stage 28 K=3 t-model pending stage 19 rerun.

11 Open Questions and Next Steps

11.1 Production model status

Item	Value	Status
Model	Stage 28: K=3, P=48, N=4,586, t-errors	Complete
$\hat{\nu}$	4.902 (95% CI [4.79, 5.01])	Complete
Lambda_a mixing	ESS=237, Rhat=1.005 (PASS)	Complete
LOO K=3 vs K=2	+1,244 $\pm$ 1,271 (direction: K=3 preferred)	Complete
Marginalized Normal ($I = 1{,}000$)	Coverage $\geq$ 85% (PASS)	Complete
Outlier study (stage 29)	All 48 features, 1,529 players, top 200 cells	Complete

11.2 Pending analyses

1. **Stage 20 rerun.** Diagonal baseline needs refitting on rebuilt data ($P = 48$, $N = 4{,}586$, K=0 stage 03). Currently running (Session 9 Block D). Once complete, the low-rank vs diagonal comparison becomes final on the rebuilt data.
2. **Stages 11–16 on stage 28.** Full ICC, loading analysis, player/season profiles for the K=3 production model. Currently these sections use the K=2 Normal model as proxy.
3. **Factor interpretation.** PC1/PC2/PC3 labels are preliminary. Draws-level PCA of $\Lambda\Lambda^\top$ gives posterior intervals on the eigenvectors, enabling credible-interval loading plots.
4. **Structured group prior.** A prior that encourages loadings to be sparse within feature groups could improve interpretability and regularisation.

11.3 Longer-term modelling

5. **Team-season effects.** A team-season effect $c_{t[n]} \in \mathbb{R}^P$ would separate manager and tactical system effects from stable individual traits. Required: a player-team mapping, which is available in the raw data.
6. **Temporal structure.** Season effects b_j are currently exchangeable. A Gaussian process or random walk over time would capture league-wide temporal trends with appropriate smoothing.
7. **Two-component mixture (deprioritised).** At $\hat{\nu} = 4.90$, the tail is moderate and well-characterised by Student-t. The mixture model (outlier fraction vs smooth heavy tail) is theoretically appealing but practically unnecessary given the current $\hat{\nu}$.